

SEO Implementation Steps — Review Document

Project Type: Next.js Website (App Router) **Purpose:** SEO Expert Review — Verify correctness of our implementation steps **Date:** June 2026

Request to Reviewer: Please review each step below and tell us:

1. Is this step correct? Any mistakes?
2. Is anything missing that should be added?
3. Is there a better approach for Next.js App Router?
4. Where should we focus first for better ranking?

STEP 1 — robots.txt

File Location: `public/robots.txt`

What we do:

- Create `robots.txt` file inside the `public` folder
- Block unnecessary pages: `/feed`, `/wp-admin/`, `/cgi-bin/`
- Do NOT block images/uploads (needed for rich results)
- Add Sitemap URL at the bottom of the file

Our robots.txt format:

```
User-agent: *  
Disallow: /feed  
Disallow: /comments/feed  
Disallow: /wp-admin/  
Disallow: /cgi-bin/  
  
Sitemap: https://yourwebsite.com/sitemap.xml
```

Questions for Expert:

- Is this format correct for a frontend-only Next.js site?
- Are there any other pages we should disallow?

STEP 2 — XML Sitemap (Dynamic Route)

File Location: `src/app/sitemap.xml/route.js`

What we do:

- Create a `sitemap.xml` folder inside the `app` directory
- Add a `route.js` file inside it
- Use `NextResponse` to dynamically generate XML sitemap
- Set `priority`, `lastmod`, and `changefreq` for each URL

Priority levels we use:

Page Type	Priority
Homepage	1.0
About / Specialities	0.9
Contact / Resources / Testimonials	0.8
Individual Article Pages	0.7

Code structure:

```
import { NextResponse } from "next/server";

export async function GET() {
  const baseUrl = "https://yourwebsite.com";
  const urls = [
    { path: "", priority: "1.0" },
    { path: "aboutus", priority: "0.9" },
    { path: "contactus", priority: "0.8" },
    { path: "allartical/laparoscopy", priority: "0.7" },
  ];

  const sitemap = `<?xml version="1.0" encoding="UTF-8"?>
<urlset xmlns="http://www.sitemaps.org/schemas/sitemap/0.9">
  ${urls.map(({ path, priority }) => `
<url>
  <loc>${baseUrl}/${path}</loc>
  <lastmod>${new Date().toISOString().split("T")[0]}</lastmod>
  <changefreq>monthly</changefreq>
  <priority>${priority}</priority>
  `)}
</urlset>`;

  return NextResponse.xml(sitemap);
}
```

```

    </url>`).join("")}
  </urlset>`;

  return new NextResponse(sitemap, {
    headers: { "Content-Type": "application/xml" },
  });
}

```

Questions for Expert:

- Is `route.js` the correct approach for dynamic sitemap in Next.js App Router?
- Should `changefreq` vary per page type (e.g., homepage = weekly)?

STEP 3 — llm.txt (AI Crawler File)

File Location: `public/llm.txt`

What we do:

- Create `llm.txt` file inside the `public` folder
- This file is for AI crawlers (ChatGPT, Claude, Gemini, Perplexity, etc.)
- It tells AI bots what the website is about, who it serves, and what content is available
- Similar to `robots.txt` but specifically for Large Language Models (LLMs)

Our llm.txt format:

```

# llm.txt
# Website: https://yourwebsite.com
# Purpose: Help AI crawlers understand this website

## About
[Brand Name] is a [describe website/business].
Located in [City], serving [target audience].

## Services / Content
- [Service/Page 1]
- [Service/Page 2]
- [Blog / Articles]

## Contact
Phone: [phone number]
Email: [email]
Address: [address]

```

```
## Important Pages
- Homepage: https://yourwebsite.com/
- About: https://yourwebsite.com/aboutus
- Contact: https://yourwebsite.com/contactus
- Blog: https://yourwebsite.com/blog
```

Why llm.txt is important:

- AI tools like ChatGPT, Perplexity, Gemini are now used for search
- If AI knows about your website, it can recommend you to users
- Future-proof SEO strategy — AI search is growing fast
- Helps AI give accurate information about your business

Questions for Expert:

- Is our llm.txt format correct?
- What extra information should we add in llm.txt for better AI visibility?
- Is llm.txt being widely adopted — should we prioritize it?

STEP 4 — Global Metadata in `layout.jsx`

File Location: `src/app/layout.jsx`

What we do:

- Set `metadataBase` URL
- Define global `title` and `description`
- Add extensive location-based `keywords` array
- Add `openGraph` : title, description, URL, siteName, image (1200×630), locale (`en_IN`), type
- Add `twitter` card: `summary_large_image`
- Set `robots`: { `index: true`, `follow: true` }
- Set `alternates.canonical`
- Add `icons` : favicon.ico, apple-touch-icon

Schema Markup (JSON-LD) also added in layout:

- Schema Type: `Physician`
- Fields: name, image, description, address, geo (lat/lng), serviceArea (30+ areas), telephone, medicalSpecialty, sameAs (social links)
- Added using: `<Script strategy="afterInteractive" type="application/ld+json">`

Questions for Expert:

- Is placing Schema Markup in `layout.jsx` correct, or should it be on specific pages only?
- Does adding too many keywords in global metadata hurt SEO (keyword stuffing)?
- Is `strategy="afterInteractive"` the right strategy for JSON-LD Schema?

STEP 5 — Page-Level Metadata (Each `page.jsx`)

File Location: `src/app/[pagename]/page.jsx`

What we do:

- Add separate `export const metadata = {}` in each page's `page.jsx`
- Define page-specific `title`, `description`, `keywords`
- Add `openGraph` with page-specific URL, title, description

Example — Contact Page:

```
export const metadata = {
  title: "Contact Dr. Shazia Waghoo | Santacruz, Mumbai",
  description: "Get in touch with Dr. Shazia Waghoo for appointments
    and medical inquiries in Santacruz, Mumbai.",
  openGraph: {
    url: "https://yourwebsite.com/contactus",
    title: "Contact Dr. Shazia Waghoo",
    description: "Get in touch for appointments and consultations.",
    siteName: "Brand Name",
  },
};
```

Pages where we add metadata:

- `/` (homepage), `/aboutus`, `/contactus`, `/resources`
- `/allartical/[articlename]` (each article page)
- `/blog/[blogname]` (each blog post)

Questions for Expert:

- Should we also add `canonical` and `robots` in each page-level metadata?
- Is there any conflict between global metadata in `layout.jsx` and page-level metadata?

STEP 6 — Meta Tags in Client Components

File Location: `src/components/[ComponentName].jsx`

What we do:

- Use `import Head from "next/head"` in client-side components
- Inside `<Head>` we add:
 - `<title>` and `<meta name="description">`
 - `<meta name="keywords">` and `<meta name="author">`
 - `<meta name="robots" content="index, follow">`
 - `<link rel="canonical">`
 - Open Graph tags (og:title, og:description, og:type, og:url, og:image)
 - Twitter Card tags
 - JSON-LD Structured Data (`<script type="application/ld+json">`) inside Head

Questions for Expert:

- In Next.js App Router, is using `next/head` in components correct, or should we always prefer `metadata` export?
- Can adding SEO tags inside multiple components cause duplicate meta tag issues?
- Should Schema markup (JSON-LD) be in one place only, or can it appear in multiple components?

STEP 7 — Google Analytics GA4

File Location: `src/components/GoogleAnalytics.jsx`

What we do:

Create `GoogleAnalytics.jsx` component:

```
"use client";
import Script from "next/script";

export default function GoogleAnalytics() {
  return (
    <>
      <Script
        src="https://www.googletagmanager.com/gtag/js?id=G-XXXXXXXXXX"
        strategy="afterInteractive"
      />
      <Script id="ga4-init" strategy="afterInteractive">
```

```

    {`
      window.dataLayer = window.dataLayer || [];
      function gtag(){dataLayer.push(arguments);}
      gtag('js', new Date());
      gtag('config', 'G-XXXXXXXXXX');
    `}
  </Script>
</>
);
}

```

Import in `layout.jsx` inside `<body>`:

```

import GoogleAnalytics from "@components/GoogleAnalytics";
<GoogleAnalytics />

```

Then replace `G-XXXXXXXXXX` with real GA4 Measurement ID.

Questions for Expert:

- Is `strategy="afterInteractive"` correct for GA4, or should we use `lazyOnload`?
- Should the GA4 Measurement ID be stored in an environment variable (`.env`)?

STEP 8 — Google Search Console Setup

What we do (manual steps):

1. Add property in Google Search Console (Domain type — recommended)
2. Verify ownership by uploading HTML verification file to `public` folder
3. Submit sitemap: enter `sitemap.xml` and click Submit
4. Monitor: Coverage, Pages (indexed / not indexed / errors)
5. Check Performance Report: clicks, impressions, keyword rankings

Questions for Expert:

- Domain property vs URL prefix — which is better for a Next.js site?
- After sitemap submission, how long should indexing take for a new site?

STEP 9 — SEO Audit Tools (Checking & Testing)

Tools we use to check SEO performance:

Tool 1 — SEOptimer (seoptimer.com)

What we do:

- Go to seoptimer.com and enter website URL
- It gives a full SEO report with grades (A, B, C, D, E) for:
 - On-Page SEO (title, meta, headings, content)
 - Links (internal/external backlinks)
 - Usability (mobile-friendly, page speed)
 - Performance (load time, page size)
 - Social (Open Graph, Twitter Card)
- We check all failing points and fix them one by one
- Re-run after fixes to verify improvement

What we check specifically:

- Missing meta descriptions
- Missing alt tags on images
- Broken links
- Page load speed
- Mobile responsiveness score
- Missing Open Graph tags

Tool 2 — Chrome Lighthouse (Built-in DevTools)

What we do:

- Open website in Chrome browser
- Press **F12** → Go to **Lighthouse** tab
- Run audit for: Performance, Accessibility, Best Practices, SEO
- Target score: **90+ in all categories**

What we check specifically:

Category	Target Score	What It Checks
Performance	90+	Page load speed, LCP, CLS, FID
Accessibility	90+	Alt text, color contrast, ARIA labels

Best Practices	90+	HTTPS, console errors, image formats
SEO	90+	Meta tags, robots, mobile-friendly

How we fix issues:

- Compress images (WebP format)
- Lazy load images and components
- Remove unused CSS/JS
- Fix missing alt attributes
- Ensure proper heading hierarchy (H1 → H2 → H3)

Questions for Expert:

- Are SEOptimizer and Lighthouse enough, or should we also use Ahrefs/SEMrush?
- What Lighthouse score is considered "good enough" to go live?
- Any other free tools we should add to our testing workflow?

STEP 10 — Google Analytics GA4 Account Setup

What we do (manual steps):

1. Go to Google Analytics → Create new account
2. Create GA4 property → Add Web data stream
3. Copy Measurement ID (`G-XXXXXXXXXX`)
4. Add to website via `GoogleAnalytics.jsx` component
5. Verify tracking via Realtime report

STEP 11 — Blog Creation on Website

File Location: `src/app/blog/[slug]/page.jsx`

What we do:

- Create a `/blog` section on the website
- Each blog post has its own page with unique URL: `/blog/[post-name]`
- Each blog post has dedicated SEO metadata:
 - Unique `title` (includes primary keyword)

- `description` (150–160 characters, includes keyword)
- `keywords` (topic-specific)
- `openGraph` with post image and URL
- `canonical` URL

Blog SEO best practices we follow:

- H1 tag = Blog post title (only one H1 per page)
- H2/H3 tags for subheadings (proper hierarchy)
- Keyword density: 1–2% (not stuffed)
- Minimum word count: 800–1200 words per post
- Internal links: Link to related service pages from blog
- Image alt text: Descriptive alt for every image
- URL structure: `/blog/keyword-rich-slug` (no spaces, lowercase, hyphens)

Example blog metadata:

```
export const metadata = {
  title: "What is Laparoscopic Surgery? Benefits & Recovery | Dr. Shazia Waghoo",
  description: "Learn about laparoscopic surgery – a minimally invasive procedure with faster recovery. Expert insights by Dr. Shazia Waghoo.",
  keywords: ["laparoscopic surgery", "minimally invasive surgery", "hernia surgery"],
  openGraph: {
    url: "https://yourwebsite.com/blog/laparoscopic-surgery-guide",
    title: "What is Laparoscopic Surgery?",
    description: "Expert guide on laparoscopic surgery by Dr. Shazia Waghoo.",
    images: [{ url: "/blog/laparoscopy-cover.jpg", width: 1200, height: 630 }],
  },
  alternates: {
    canonical: "https://yourwebsite.com/blog/laparoscopic-surgery-guide",
  },
};
```

Questions for Expert:

- How frequently should we publish blog posts for good SEO impact?
- Should blog posts be indexed immediately or after some minimum word count?
- Is internal linking from blog to service pages important for ranking?

STEP 12 — Social Media SEO (Off-Page SEO)

Platforms we use:

YouTube

What we do:

- Create YouTube channel for the brand
- Upload educational/informational videos
- Add website URL in video description and channel bio
- Use keyword-rich video titles and descriptions
- Add website link in YouTube channel header

SEO benefit:

- Google shows YouTube videos in search results (Video Rich Results)
- Backlink from YouTube to website improves domain authority
- Brand visibility increases

Instagram

What we do:

- Create Instagram business profile
- Add website URL in bio
- Post content related to services (images, reels, stories)
- Use relevant hashtags with local + service keywords
- Example hashtags: `#herniaspecialist #laparoscopicsurgeon #mumbaidoctor`
- Add location tags to every post (local SEO benefit)
- Share blog post links in stories with "Link" sticker

SEO benefit:

- Profile appears in Google search for brand name
- Social signals help in off-page SEO
- Drives traffic to website

Facebook

What we do:

- Create Facebook Business Page
- Add website URL in page information
- Share blog posts and service pages on Facebook
- Post content regularly (at least 3–4 times per week)

- Create Facebook events for camps/health seminars
- Enable reviews on Facebook page

SEO benefit:

- Facebook page ranks on Google for brand name searches
- Backlink from Facebook to website
- Reviews on Facebook improve brand trust and local SEO

Questions for Expert:

- Which social media platform is most important for local business SEO?
- How many posts per week is ideal for SEO benefit on social media?
- Should we embed social media feed on the website? Any SEO benefit?

Complete Implementation Summary

#	Step	Type	Status
1	<code>public/robots.txt</code>	Technical SEO	Done
2	<code>app/sitemap.xml/route.js</code>	Technical SEO	Done
3	<code>public/llm.txt</code>	AI SEO	Done
4	Global metadata in <code>layout.jsx</code>	On-Page SEO	Done
5	Page-level metadata in <code>page.jsx</code>	On-Page SEO	Done
6	Meta tags in components	On-Page SEO	Done
7	Google Analytics GA4	Tracking	Done
8	Google Search Console	Technical SEO	Done
9	SEOptimer + Lighthouse audits	SEO Testing	Done
10	GA4 account setup	Tracking	Done
11	Blog creation with SEO metadata	On-Page SEO	In Progress
12	Social Media (YouTube, Instagram, Facebook)	Off-Page SEO	In Progress

SEO Types We Cover

SEO Type	What We Cover
On-Page SEO	Title, meta description, keywords, Schema markup, blog content
Technical SEO	robots.txt, sitemap, canonical, llm.txt, mobile-friendly
Off-Page SEO	Google Search Console, Social media (YouTube, Instagram, Facebook)
AI SEO	llm.txt for AI crawlers (ChatGPT, Perplexity, Gemini)
SEO Testing	SEOptimer audit, Chrome Lighthouse (Performance, SEO, Accessibility)